Siyabonga Makhubu

AI Engineer Intern

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Professional Summary

AI Engineer Intern with hands-on experience developing AI-powered applications, LLM-based solutions, and intelligent automation workflows. Skilled in Python, TypeScript, JavaScript, C#, SQL, REST APIs, and backend development, with practical experience in Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), prompt engineering, AI agents, LangChain, MCP, and n8n automation. Experienced in document processing, embeddings, semantic search, BM25, hybrid retrieval, reranking, LLM integration, and AI evaluation using DeepEval. Familiar with machine learning workflows, model fine-tuning, application testing, and evaluating AI reliability, including faithfulness, relevance, and evidence grounding. Project experience includes AI-powered job matching, voice-based interview preparation, MCP integrations, Multichoice RAG Agent and Microsoft Fabric lakehouse solutions. Previously worked as a Revenue Assurance Graduate at MultiChoice Group, applying SQL, Power BI, discrepancy analysis, and KPI/KCI monitoring to support data quality, revenue accuracy, and loss prevention. Passionate about building secure, reliable, scalable, and practical AI solutions that address real-world challenges.

Technical Skills

- **AI and Machine Learning:** Machine learning, deep learning, natural language processing, computer vision, model training, feature engineering, model evaluation, Scikit-learn, TensorFlow, PyTorch, Hugging Face Transformers, Pandas, NumPy.
- **Generative AI and LLM Engineering:** Large Language Models (LLMs), prompt engineering, Retrieval-Augmented Generation (RAG), embeddings, vector search, AI agents, Model Context Protocol (MCP), model fine-tuning concepts, LLM integration, DeepEval, AI evaluation workflows, n8n automation, LangChain, Groq, OpenAI API, and Mistral AI.
- **Software Development:** Frontend and backend development, API design, REST APIs, Python, TypeScript, JavaScript, C#, SQL, HTML, CSS, FastAPI, ASP.NET Core Web API, LangChain, Express.js, React, Node.js, Pydantic, and Vite. Object-oriented programming, data structures and algorithms
- **Data and Databases:** SQL, data preprocessing, data pipelines, ETL, vector search MongoDB Atlas, OneLake, Data WareHouse, LakeHouse

- **MLOps and Cloud:** Docker, CI/CD, model serving.
- **Engineering Practices:** Git, version control, automation testing(Playwright, Selenium, Pytest), debugging, code review, documentation, agile collaboration, reproducible development

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AI and Software Projects

Siya's AI Job Assistance — AI Engineering Project

GitHub repo: Siyabonga-Makhubu/AI_Job_Search_Assistant

- Built an AI-powered job assistance system using Python, TypeScript, Node.js, MongoDB Atlas, and REST APIs to support resume-based job discovery, candidate-job matching, and interview preparation.
- Implemented an LLM-powered retrieval workflow using document processing, embeddings, semantic search, BM25 keyword retrieval, hybrid search, reranking, and Groq-hosted language models to generate evidence-grounded job-matching insights.
- Developed API and retrieval-testing workflows and incorporated DeepEval evaluation to assess LLM response quality, with automated UI and API testing using Playwright as part of the testing approach.
- **Technology Stack:** Python, TypeScript, Node.js, Express.js, MongoDB Atlas, REST APIs, Groq, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), embeddings, semantic search, BM25, hybrid retrieval, reranking, DeepEval, Playwright, Model Context Protocol (MCP).

DStv Customer Support RAG — AI Engineering Project

Github repo: Siyabonga-Makhubu/Multichoice_RAG

- Built a customer-support application using Python, FastAPI, TF-IDF, and n8n to answer customer queries about account details, billing, package entitlements, support tickets, and upgrade options using retrieved information.
- Implemented a Retrieval-Augmented Generation (RAG) workflow with customer-specific document chunking, TF-IDF similarity search, customer-ID filtering, official website ingestion, and context-grounded LLM response generation to improve the relevance and reliability of customer-support answers.
- Developed API-based validation workflows integrating two independent LLM validators through n8n, withholding generated answers if either validator rejects them; implemented input validation, HTTPS domain allowlisting, TLS certificate verification, and explicit insufficient-information responses to support security and reliability.
- **Technology Stack:** Python, FastAPI, Retrieval-Augmented Generation (RAG), TF-IDF, scikit-learn, n8n, REST APIs, webhooks, LLM integration, document chunking, similarity search, pytest, input validation, HTTPS, TLS certificate verification.
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Keamogetswe Moagi — Chef Portfolio Website

Live Portfolio: https://siyabonga-makhubu.github.io/Keamo_Chef_Portfolio/

GitHub repo: Siyabonga-Makhubu/Keamo_Chef_Portfolio

- Developed a responsive, multi-page chef portfolio using HTML, CSS, vanilla JavaScript, and Tailwind CSS to showcase culinary skills, qualifications, professional experience, food photography, and contact information.
- Implemented an interactive enquiry form with client-side validation and WhatsApp integration, allowing visitors to prepare and send enquiries directly through WhatsApp without requiring a portfolio backend.
- Automated browser testing using Python, pytest, Playwright, and Selenium to support UI validation and functional testing, and configured GitHub Actions with GitHub Pages for automated static-site deployment. Used Git LFS to manage selected large media assets.
- **Technologies:** HTML, CSS, JavaScript, Tailwind CSS, Python, GitHub Actions, GitHub Pages, Playwright, Selenium

Siyabonga Makhubu — Personal Portfolio Website

Live Portfolio: https://siyabonga-makhubu.github.io/Siyabonga-Portfolio/

GitHub repo: Siyabonga-Makhubu/Keamo_Chef_Portfolio

- Developed a responsive personal portfolio website using HTML, CSS, and JavaScript to showcase technical skills, education, certifications, professional experience, and software development projects.
- Implemented interactive UI components, animated sections, a skills showcase, education timeline, project gallery with GitHub links, social media links, and contact form integration using EmailJS.
- Showcased a range of software and AI engineering projects, including an ASP.NET Core survey application, legal practice management system, DStv Customer Support RAG application, and AI Job Search Assistant with resume-based retrieval and MCP integration.
- Deployed the portfolio using GitHub Pages and managed source code and project updates through Git and GitHub, providing recruiters and collaborators with a central platform to explore my technical capabilities and development experience.
- Technologies: HTML5, CSS3, JavaScript, EmailJS, Git, GitHub, GitHub Pages

Auto-Tracker — Legal Practice Management System

- Developed a single-page legal practice management prototype using HTML, Tailwind CSS, and vanilla JavaScript to centralise time tracking, client and matter management, invoicing, trust accounting, and productivity reporting.
- Implemented a time-tracking interface with six-minute billing increments, manual time-entry functionality, and simulated activity detection to demonstrate how legal

professionals could capture billable work and reduce manual administration.
- Built invoicing and reporting features to generate invoices from approved time entries, track client trust balances, and visualise time allocation, utilisation, and efficiency metrics using ApexCharts.
- Integrated SheetJS-based Excel export and designed reporting workflows for monthly summaries, with PDF export functionality included in the prototype's intended feature set.
- **Project Status:** Prototype / demo application. Activity detection is simulated, and backend integration and real-world API connections remain planned.
- **Technologies:** HTML5, Tailwind CSS, JavaScript, ApexCharts, Lucide Icons, SheetJS

GitHub repo: Siyabonga-Makhubu/Auto-Tracker-legal-management-system: Auto Tracker legal management system

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Professional Experience

AI Engineer Intern | Indsafri | Honeydew, Gauteng, South Africa

1 July 2026 – Present

- Develop AI-powered applications using Large Language Models (LLMs), Generative AI, prompt engineering, and retrieval-based techniques to address practical use cases.
- Implement AI workflows involving document processing, embeddings, semantic search, hybrid retrieval, and reranking to support evidence-grounded responses and intelligent information retrieval.
- Integrate LLMs and external services into software applications using REST APIs, LangChain, TypeScript, Python, and Model Context Protocol (MCP).
- Apply AI evaluation and testing practices using DeepEval, pytest, and API testing tools to assess response quality, reliability, and application behaviour.
- Develop automated UI and end-to-end testing workflows using Playwright and Selenium, while exploring performance testing with Locust.
- Use GitHub Copilot to support code development, debugging, documentation, and problem-solving while reviewing AI-generated code for correctness and maintainability.
- Apply software engineering principles, including modular design, input validation, error handling, and API integration, when developing maintainable AI applications.

Key Skills: AI Engineering, LLMs, Generative AI, Prompt Engineering, RAG, Embeddings, MCP, LangChain, DeepEval, REST APIs, Python, TypeScript, MongoDB, n8n, GitHub Copilot, pytest, Playwright, Selenium, Locust, Postman.

Revenue Assurance Graduate | MultiChoice Group | Randburg, Gauteng, South Africa

March 2023 – May 2025

- Developed and executed SQL scripts to identify discrepancies between billing and provisioning systems, supporting revenue assurance investigations and revenue leakage prevention.
- Analysed billing, provisioning, and operational data using Oracle SQL Developer, MySQL, and Microsoft Excel to identify anomalies and support data-driven decision-making.
- Built and maintained Power BI dashboards to monitor discrepancies, track trends, and communicate revenue assurance performance to stakeholders.
- Monitored Key Control Indicators (KCIs) and Key Performance Indicators (KPIs) to identify potential risks affecting revenue accuracy and operational performance.
- Collaborated with internal stakeholders, external vendors, and consultants to investigate system discrepancies and support revenue assurance initiatives.
- Contributed to process improvement activities aimed at strengthening data validation, improving reporting efficiency, and reducing potential revenue leakage.
- Prepared analytical reports and presented business insights to support risk identification, operational monitoring, and revenue recovery activities.

Key Skills: SQL, Oracle SQL Developer, MySQL, Microsoft Excel, Power BI, Data Analysis, Revenue Assurance, Data Validation, KPI/KCI Monitoring, Reporting, Stakeholder Collaboration.

Education

Bachelor of Science (BSc) in Computer Science and Information Technology

University of KwaZulu-Natal | Durban, KwaZulu-Natal, South Africa
Completed: May 2023

- Developed a strong foundation in software development, programming, database management, algorithms, and computational problem-solving.
- Gained academic experience with Python, Java, C#, JavaScript, HTML, CSS, SQL, and Linux.
- Completed practical programming assignments and projects focused on applying computing concepts to real-world problems.

National Senior Certificate (Matric)

Injabulo Combined School | Piet Retief, Mpumalanga, South Africa
Completed: December 2017

- Completed secondary education, developing foundational analytical, communication, teamwork, and problem-solving skills.

Certifications and Achievements

- **Certification:** Full Stack Development | IT Varsity | Credential Id: AOTYAA072025 | Issued by July 2025
- **Certification:** Business Data Analysis | Udemy | Credential Id: UC-b3f903f0-1959-4342-903c-5f70f74005e | Issued by March 2025